

CSCI 1101 Introduction to Computer Science



INTRODUCTION TO BOOLEAN ALGEBRA

Circuits and Boolean Algebra

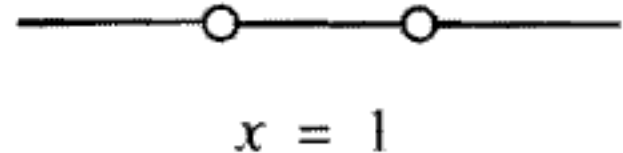
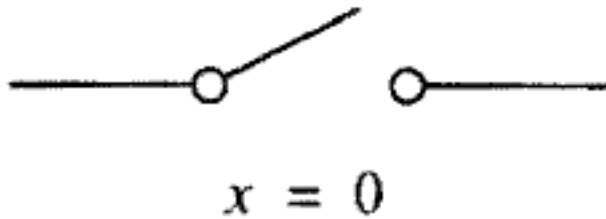
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- Computers are constructed from **logic circuits**
 - Logic circuits perform operations on digital signals
 - Usually implemented as circuits where signal values are restricted to a few values (e.g., binary circuits)
- But first, a short intro to **Boolean algebra**

Binary Circuits

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- Circuit with 2 possible states, open ($x = 0$) or closed ($x = 1$), where x is the input variable that controls the switch



Boolean Algebra

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- Named for George Boole, who in 1849 introduced a scheme for describing in algebra the processes involved in logical reasoning
- In 1930's, Claude Shannon showed that Boolean algebra provided a means of describing circuits built with switches
- Provides foundation for much of modern technology

Fundamental Operations

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- Boolean algebra works with variables that can have only 2 values – **true** or **false** (1 or 0, respectively)
- The fundamental operations of Boolean algebra are AND, OR, and NOT

AND and OR Operations

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- An AND function of variables $x_1, x_2, \dots, x_n$ has the value 1 only if all n variables are equal to 1.
 - AND is like multiplication
- An OR function of variables $x_1, x_2, \dots, x_n$ has the value 1 if at least one, or more, of the n variables are equal to 1
 - OR is like addition

NOT Operation

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- NOT takes one variable/argument/input
- NOT is true if the input variable is false
 - NOT (true) = false
 - NOT (false) = true
 - We'll see another way to represent this in a minute

Truth Tables

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- Used to define logic operations

x_1	x_2	$x_1 \cdot x_2$	$x_1 + x_2$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1
		AND	OR

Truth table for AND and OR operations with 2 inputs, x_1 and x_2

Extending AND and OR

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- AND and OR can be extended to n variables

x_1	x_2	x_3	$x_1 \cdot x_2 \cdot x_3$	$x_1 + x_2 + x_3$
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Truth table for
AND and OR
operations with 3
inputs

*What would the
truth table for NOT
look like?*

Axioms of Boolean Algebra

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- Boolean algebra is based on set of rules that are derived from a small number of basic assumptions – **axioms**
- Basic assumption: Boolean algebra, B , involves elements that take on one of two values, 0 and 1.
- The following axioms are assumed to be true...

Axioms of Boolean Algebra

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$$1a. \quad 0 \cdot 0 = 0$$

$$1b. \quad 1 + 1 = 1$$

$$2a. \quad 1 \cdot 1 = 1$$

$$2b. \quad 0 + 0 = 0$$

$$3a. \quad 0 \cdot 1 = 1 \cdot 0 = 0$$

$$3b. \quad 1 + 0 = 0 + 1 = 1$$

$$4a. \quad \text{If } x = 0, \text{ then } \bar{x} = 1$$

$$4b. \quad \text{If } x = 1, \text{ then } \bar{x} = 0$$

Single Variable Rules (Theorems)

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- From the axioms, we define rules (theorems) for dealing with single variables
- We can prove by *perfect induction* (substitute $x = 0$ and $x = 1$ in each expression and use axioms)

$$5a. \quad x \cdot 0 = 0$$

$$5b. \quad x + 1 = 1$$

$$6a. \quad x \cdot 1 = x$$

$$6b. \quad x + 0 = x$$

$$7a. \quad x \cdot x = x$$

$$7b. \quad x + x = x$$

$$8a. \quad x \cdot \bar{x} = 0$$

$$8b. \quad x + \bar{x} = 1$$

$$9. \quad \bar{\bar{x}} = x$$

Principle of Duality in Boolean Algebra

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- Given a logic expression, we get its *dual* by replacing all $+$ with $\cdot$ (and *vice versa*) and by replacing 0s with 1s and 1s with 0s.
- The dual of any true statement, axiom or theorem, in Boolean algebra is also a true statement
- This means that at least 2 different ways exist to express every logic function

Two- and Three Variable Properties

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- We often need to deal with a number of variables
- Define some 2- and 3-variable identities
- Give the dual of each identity
- Often referred to as *properties*
- If x , y , and z are variables in B , the following properties hold...

Properties

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10a. $x \cdot y = y \cdot x$ *Commutative*

10b. $x + y = y + x$

11a. $x \cdot (y \cdot z) = (x \cdot y) \cdot z$ *Associative*

11b. $x + (y + z) = (x + y) + z$

12a. $x \cdot (y + z) = x \cdot y + x \cdot z$ *Distributive*

12b. $x + y \cdot z = (x + y) \cdot (x + z)$

13a. $x + x \cdot y = x$ *Absorption*

13b. $x \cdot (x + y) = x$

14a. $x \cdot y + x \cdot \bar{y} = x$ *Combining*

14b. $(x + y) \cdot (x + \bar{y}) = x$

Properties

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$$15a. \quad \overline{x \cdot y} = \bar{x} + \bar{y}$$

DeMorgan's theorem

$$15b. \quad \overline{\bar{x} + \bar{y}} = x \cdot y$$

$$16a. \quad x + \bar{x} \cdot y = x + y$$

$$16b. \quad x \cdot (\bar{x} + y) = x \cdot y$$

$$17a. \quad x \cdot y + y \cdot z + \bar{x} \cdot z = x \cdot y + \bar{x} \cdot z$$

Consensus

$$17b. \quad (x + y) \cdot (y + z) \cdot (\bar{x} + z) = (x + y) \cdot (\bar{x} + z)$$

Properties

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- We can prove by perfect induction (truth tables), algebraic manipulation, or Venn diagrams
- Example: proving DeMorgan's theorem using truth table

x	y	$x \cdot y$	$\overline{x \cdot y}$	$\bar{x}$	$\bar{y}$	$\bar{x} + \bar{y}$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

$\underbrace{\hspace{10em}}$
LHS

$\underbrace{\hspace{10em}}$
RHS

Venn Diagrams

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- Using truth tables to verify theorems and properties can be tedious, and may not be conceptually informative
- Venn diagrams are a tool that provides for more intuitive understanding of how expressions may be equivalent
 - Graphical illustration of set algebra
 - We're not going to talk about them.

Equivalent Notations

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The following notations are equivalent:

X' same as $\bar{X}$

$(X + Y)'$ same as $\overline{X + Y}$

Other Notation

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- Using the same symbols for arithmetic and logic can be confusing
- We can use the following notation:
 - AND $\wedge$, *e.g.*, $x_1 \wedge x_2$
 - OR $\vee$, *e.g.* $x_1 \vee x_2$

Precedence of Operations

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- In the absence of parentheses, a logic expression is evaluated in the following order:
 1. NOT
 2. AND
 3. OR